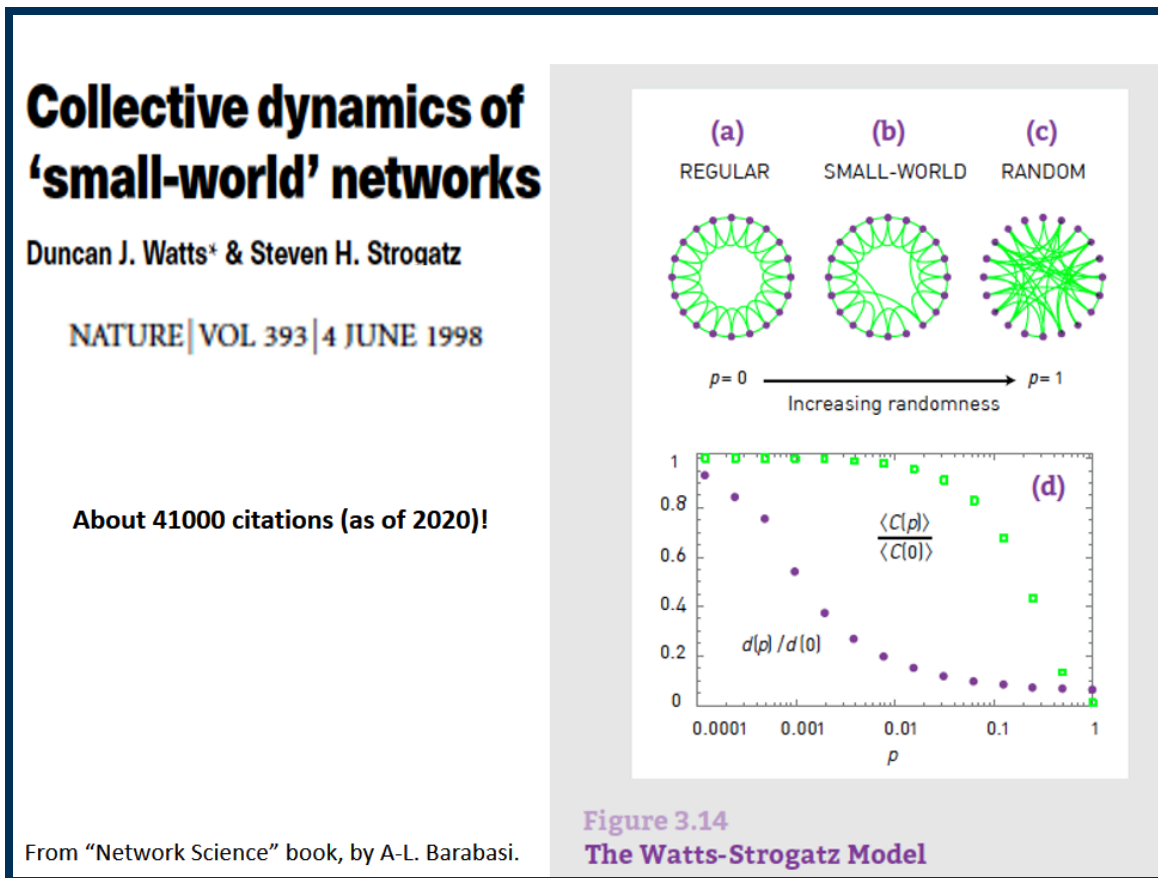


L1: The Birth of Network Science

The birth of Network Science took place back in 1998 or 1999, with the publication of two very influential research papers.



The **first** was the discovery by Watts and Strogatz of the Small-World property in real-world networks. Roughly speaking, this means that most node-pairs are close to each other, only within a small number of hops. You may have heard the term “*six degrees of separation*”, in the context of social networks, meaning that most people are connected with each other through a path of 6 (or so) acquaintances.

REPORT

Emergence of Scaling in Random Networks

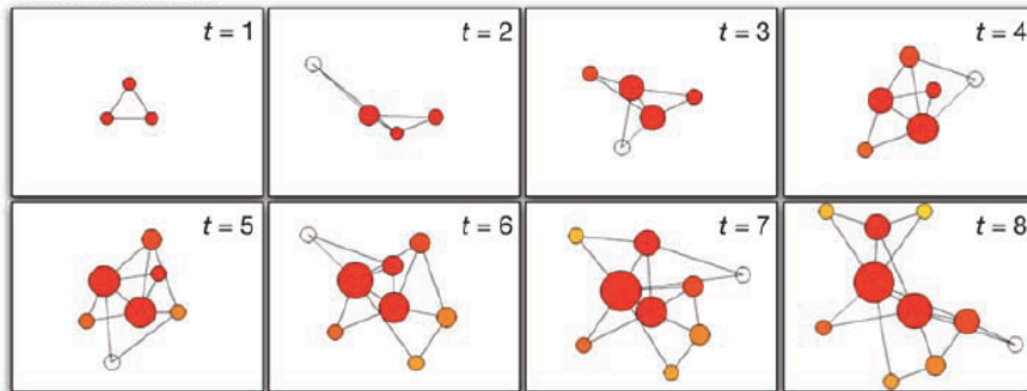
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About 35000 citations (as of 2020)!

Science 15 Oct 1999:
Vol. 286, Issue 5439, pp. 509-512
DOI: 10.1126/science.286.5439.509

Scale-Free Model



<https://science.sciencemag.org/content/325/5939/412>

A **second** influential paper was published in 1999 by two physicists, Barabási and Albert.

That paper showed that real-world networks are “*Scale Free*”. This means that the number of connections that a node has is highly skewed: most nodes have a very small number of connections but there are few nodes, referred to as hubs, that have a much larger number of connections. Mathematically speaking, the number of connections per node follows a power-law distribution – something that we will discuss extensively later in this course.

Barabási and Albert explained this general phenomenon based on a “*rich get richer*” property. As a network gradually grows, new nodes prefer to create links to more well-connected existing nodes, and so the latter become increasingly more powerful in terms of connectivity. This is referred to as “*preferential attachment*” – and we will study it in detail later.

Source Links

- [Collective dynamics of ‘small-world’ networks](https://www.nature.com/articles/30918) ➡
(<https://www.nature.com/articles/30918>)

- Emergence of Scaling in Random Networks 
(<https://science.sciencemag.org/content/286/5439/509>)